Class Adm No

Candidate Name: ________________________________

Class	Adm No

H

millennia institute

2021 Prelim Examination
Pre-university 3

H2 Biology **9744/01**

Paper 1 Multiple Choice **20 September 2021**
1 hour

Additional Materials: Optical Answer Sheet

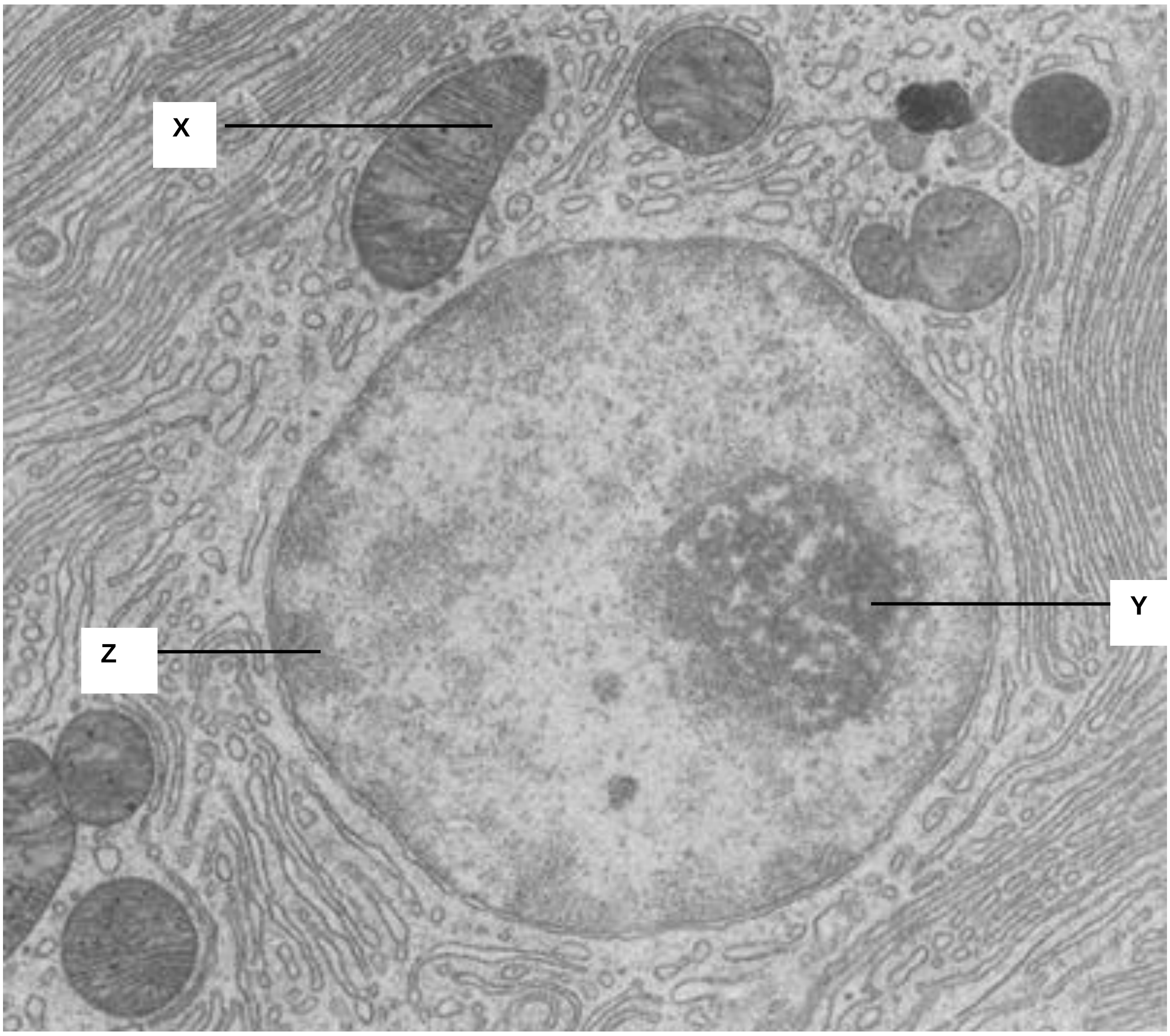
READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, Adm No. and class on all the papers you hand in.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**.
Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Each correct answer will score one mark. A mark will not be deducted for wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

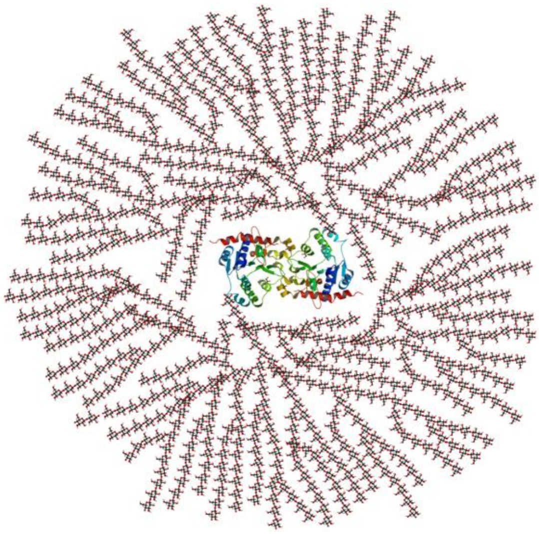
1. The electron micrograph below shows the structures found in a cell.



Which of the following statement best describe the structures **X**, **Y** and **Z**?

	X	Y	Z
A	Contains reduced NAD and reduced FAD	Transcription of gene coding for ribosomal RNA	Contains heterochromatin which is transcriptionally inactive
B	Involved in oxidative phosphorylation	Transcription of gene coding for ribosomal protein	Contains heterochromatin which is transcriptionally active
C	Contains reduced NAD and reduced FAD	Involved in assembly of ribosomal subunits	Contains euchromatin which is transcriptionally inactive
D	Involved in oxidative phosphorylation	Involved in synthesis of ribosomal subunits	Contains euchromatin which is transcriptionally active

2. Glycogen is a storage polysaccharide found in animal cells. The following figure shows a glycogen granule that is composed of glycogenin protein in the core of the granule.



Which statements correctly describes the structure of glycogen granule?

1. The bonds found in glycogen granules are α-(1,4)-glycosidic bonds and α-(1,6)-glycosidic bonds only.
2. Glycogen is a branched molecule and the glycosidic bonds can be hydrolysed easily to release α-glucose.
3. The presence of α-(1,6)-glycosidic bonds helps glycogen to fold into a compact shape.
4. Glycogenin is involved in the polymerisation of α-glucose to form the polymer.

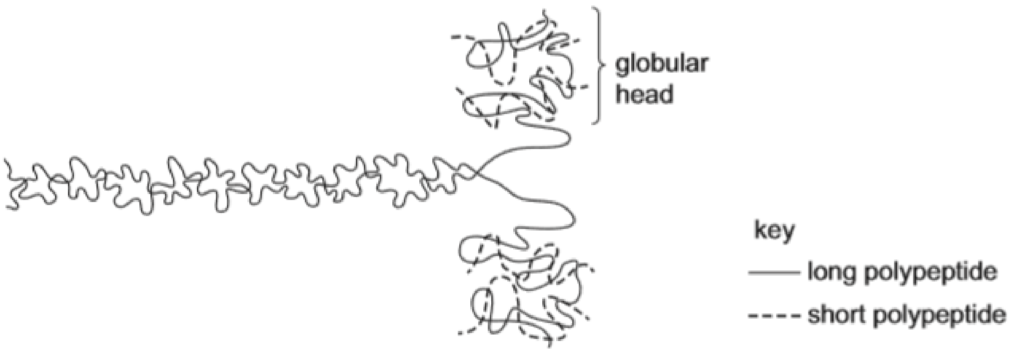
A 1 and 2 only

B 1 and 3 only

C 2 and 4 only

D 3 and 4 only

3. The diagram shows a protein molecule.



Two long polypeptides each form α-helices for most of their length and twist together into a fibre. At one of end of the fibre, each of the long polypeptide coils into a globular head with another 2 short polypeptides. There are two globular heads in each protein molecule.

Which describes the protein molecule?

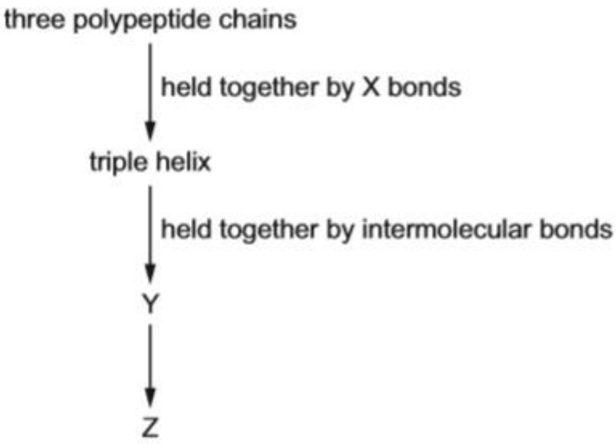
A The primary structure of the protein molecule only comprises the unique sequence of amino acids in the long polypeptide chain.

B The secondary structure of the protein molecule composed of α-helices formed by the short and long polypeptide chains.

C The tertiary structure of the protein refers to the globular head of long and short polypeptide chains.

D The quaternary structure of the protein has 6 polypeptide chains.

4. Formation of collagen is a multi-step process. The flow chart shows part of the steps in the formation of collagen.

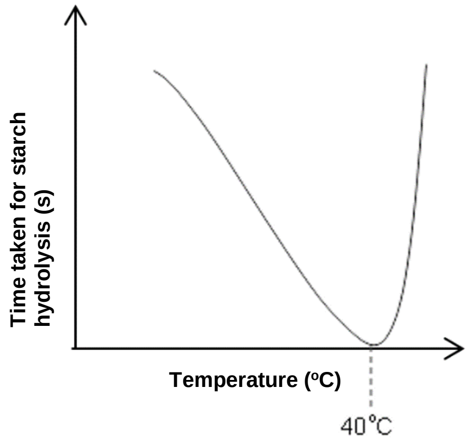


Which row correctly identifies **X**, **Y** and **Z**?

	X	**Y**	**Z**
A	covalent	fibres	fibrils
B	covalent	fibrils	fibres
C	hydrogen	fibres	fibrils
D	hydrogen	fibrils	fibres

5. A student investigated the effect of temperature on starch hydrolysis. The progress of the breakdown of starch by amylase complex is monitored by the disappearance of starch using iodine solution. At every 15 minutes interval, a drop of reaction mixture is tested with iodine solution to determine the presence of starch.

The time taken for starch hydrolysis at different temperatures is plotted on the graph shown below.



What can be correctly concluded from the investigation?

A The time taken for starch hydrolysis is determined as the time taken for iodine solution to turn blue-black.

B At optimal temperature, the time taken for iodine solution to change colour is the shortest.

C Formation of glucose will turn iodine solution to blue-black, representing time taken for starch hydrolysis.

D At around 40°C, iodine solution will remain yellow when added to the reaction mixture very quickly after the start of experiment.

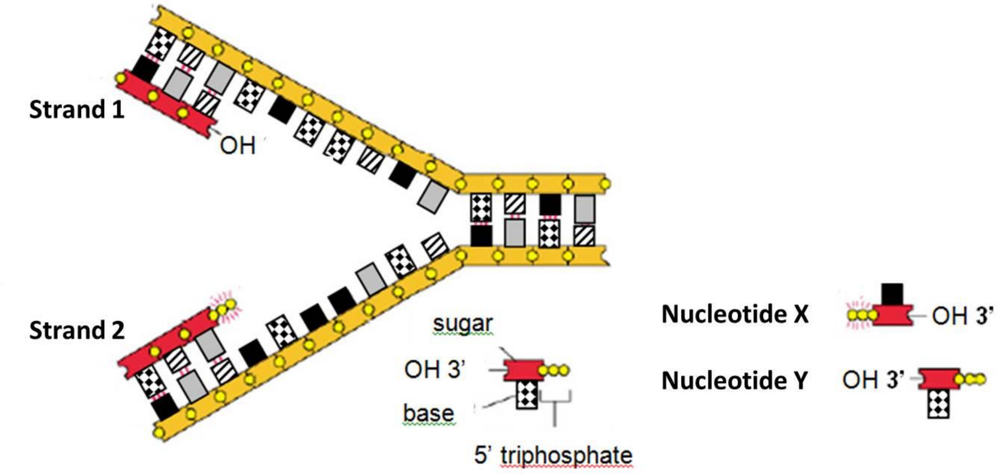
6. During an enzyme-catalysed reaction, these events occur.

1 substrate collides with enzyme
2 enzyme-substrate complex forms
3 enzyme releases product

Which row describes one possible set of processes that happen at events **1**, **2** and **3**?

	1	**2**	**3**
A	conformational changes to aid temporary bonding	conformational changes to break bonds	bonds stressed to reduce activation energy
B	conformational changes to aid temporary bonding	bonds stressed to reduce activation energy	conformational changes to break bonds
C	conformational changes to break bonds	conformational changes to aid temporary bonding	bonds stressed to reduce activation energy
D	bonds stressed to reduce activation energy	conformational changes to break bonds	conformational changes to aid temporary bonding

7. DNA replication is illustrated in the figure below.



Which of the following correctly describes the addition of the next nucleotide(s) to the DNA strands undergoing replication?

A Nucleotide **X** will be added to the leading strand, which is strand 1.
B Nucleotide **Y** will be added to the leading strand, which is strand 1.
C Nucleotide **X** will be added to the lagging strand, which is strand 2.
D Nucleotide **Y** will be added to the leading strand, which is strand 2.

8. A student made the follow statements comparing mRNA with DNA from a eukaryote.

1. A body cell has two copies of DNA coding for a particular protein but it can have thousands of copies of identical mature mRNA coding for the same protein.
2. Each DNA molecule in human cells codes for hundreds of proteins, but each mature mRNA molecule codes for the translation of only one protein.
3. Every mature mRNA is identical in length to the gene due to complementary base pairing.
4. A silent mutation will only affect the DNA nucleotide sequence, but will not change the mature mRNA nucleotide sequence.

How many statement(s) is/are true?

A 1
B 2
C 3
D 4

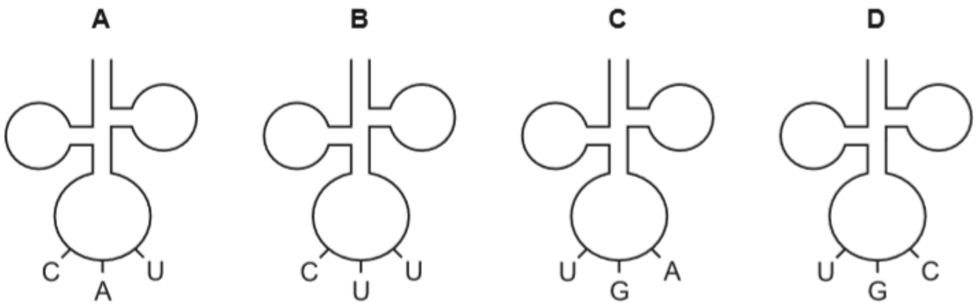
9. In the sickle cell haemoglobin, the amino acid sequence is altered. Part of the amino acid sequence in normal and sickle cell haemoglobin are shown.

Normal haemoglobin: thr-pro-glu-glu
Sickle cell haemoglobin: thr-pro-val-glu

Possible mRNA codons for these amino acids are shown below:

Amino acid	mRNA codons
Glutamine (glu)	GAA GAG
Threonine (thr)	ACU ACC
Proline (pro)	CCU CCC
Valine (val)	GUA GUG

Which tRNA is not involved in the formation of this part of the sickle haemoglobin?



10. Down's syndrome results from chromosomal aberration of chromosome 21. This can occur in several ways.

1 During gamete formation the homologous pair of chromosomes of chromosome 21 fail to separate. A gamete with two copies of chromosome 21 is fertilised by a gamete with one copy of chromosome 21.

2 During early development of the embryo, chromatids of a chromosome 21 fail to separate.

3 Translocation of part of chromosome 21 onto chromosome 14 during the development of an embryo. During gamete formation, chromosomes separate normally but may contain an extra chromosome 21.

Some individuals have a form of Down's syndrome called mosaicism. In these cases some of the body cells have the usual two copies of chromosome 21 and other cells have three copies of chromosome 21.

Which of the way(s) could explain how the mosaicism form of Down's syndrome occurs?

A 2 only

B 3 only

C 1 and 2 only

D 1 and 3 only

11. Many genes in cancer cells can be upregulated or downregulated without the gene itself being mutated. In colon cancer, 600 to 800 genes are silenced via DNA methylation of promoter region. In breast cancer, a crucial gene *BRCA1* is usually mutated, however it has been found that *BRCA1* promoter region are also commonly methylated in breast cancer patients.

Expression of *BRCA1* has also been observed to be reduced in some breast cancer patients from the over-expression of a microRNA. The microRNA is complementary to the *BRCA1* mRNA, such that upon binding, it forms a double stranded RNA complex which then signals for degradation of the complex.

Which of the following statements are true?

1 Methylation of promoter prevents binding of ribosome

2 For tumour suppressor genes to be silenced, DNA methylation must occur on the promoter of both alleles

3 microRNA regulates gene expression at the transcriptional level

4 *BRCA1* is a tumour suppressor gene

5 microRNA complementary to oncogenes mRNAs can be a potential drug to reduce cancer cell proliferation

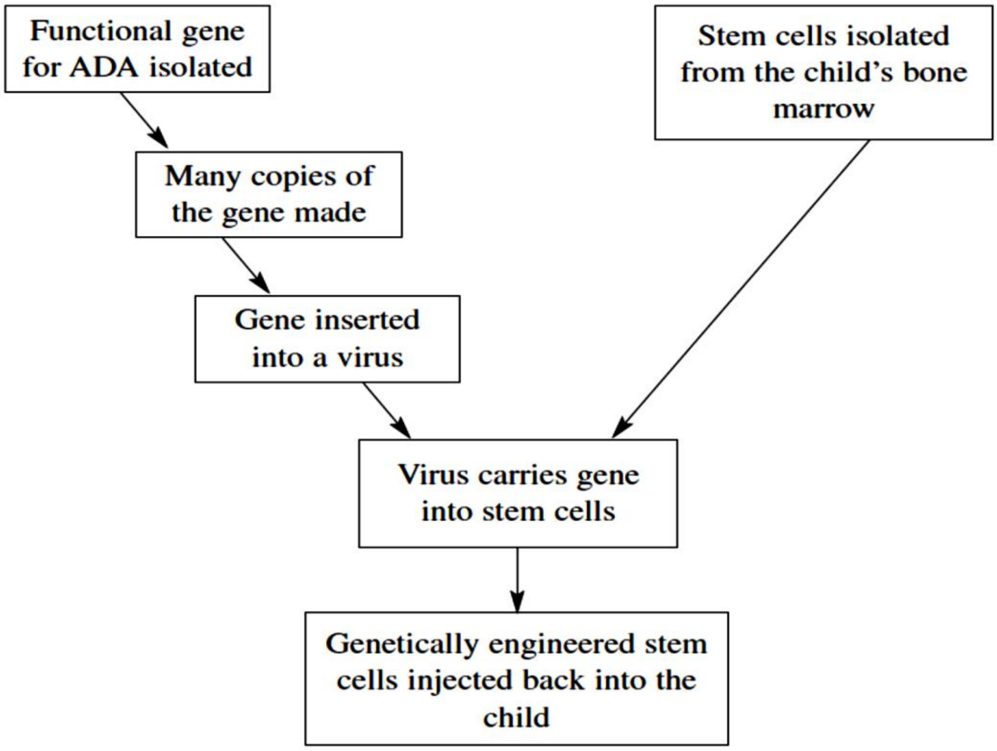
A 1 and 3 only

B 2 and 5 only

C 2, 4 and 5 only

D 1, 2, 3 and 4 only

12. Children with severe combined immunodeficiency disorder (SCID) cannot produce the many types of white blood cells that fight infections. This is because they do not have the functional gene to make the enzyme ADA. Some children with SCID have been treated with stem cells. The treatment used with the children is described in the flowchart.



Which of the following statements explains why stem cells can be used in the treatment of SCID?

1 They can divide mitotically to replace existing cells.

2 Due to their pluripotent nature, they have the ability to form only certain types of white blood cells that restores the ability to fight infection.

3 As the stem cells are from the child's own cells, there is little risk of rejection.

4 They possess a unique set of genome to allow for multipotency.

A 1 and 2 only

B 1 and 3 only

C 2 and 4 only

D 3 and 4 only

13. In human, some stem cells divide and give rise to phagocytes.

Where in the human body do these stem cells divide?

1 Blood
2 Bone marrow
3 Lymph nodes

A 2 only

B 3 only

C 1 and 3 only

D 1, 2 and 3

14. A scientist starts her experiment with an ampicillin-resistant strain of bacteria which has a non-functional *lac Z* gene. She adds F plasmids which contain a functional copy of *lac Z* gene to a tube of such bacteria (+). A control tube (-) with the same bacteria but no plasmid was prepared.

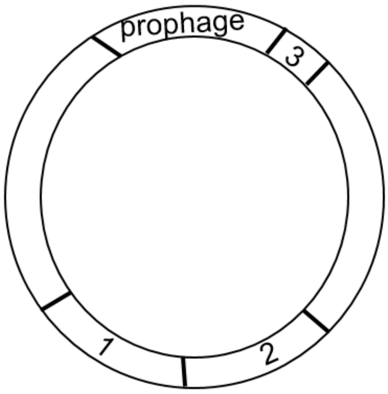
Both tubes are incubated under the appropriate conditions for growth and plasmid uptake. The scientist then spreads a sample of each bacterial culture (+ and -) on each of the three types of plates indicated below.

plate A: glucose medium
plate B: glucose medium with ampicillin
plate C: lactose medium with ampicillin

Which row correctly identifies the plates with bacterial growth (✓) or no growth (✗)?

	bacterial strain with added plasmid (+)			bacterial strain with no plasmid (-)		
	plate A	plate B	plate C	plate A	plate B	plate C
A	✓	✓	✓	✓	✓	✓
B	✓	✓	✓	✓	✓	✗
C	✓	✓	✓	✓	✗	✗
D	✓	✓	✗	✓	✓	✗

15. The figure below shows the chromosome from phage-infected bacteria. These bacteria carry the genes that code for enzymes required for synthesis of histidine (*His*$^+$), arginine (*Arg*$^+$), and methionine (*Met*$^+$).



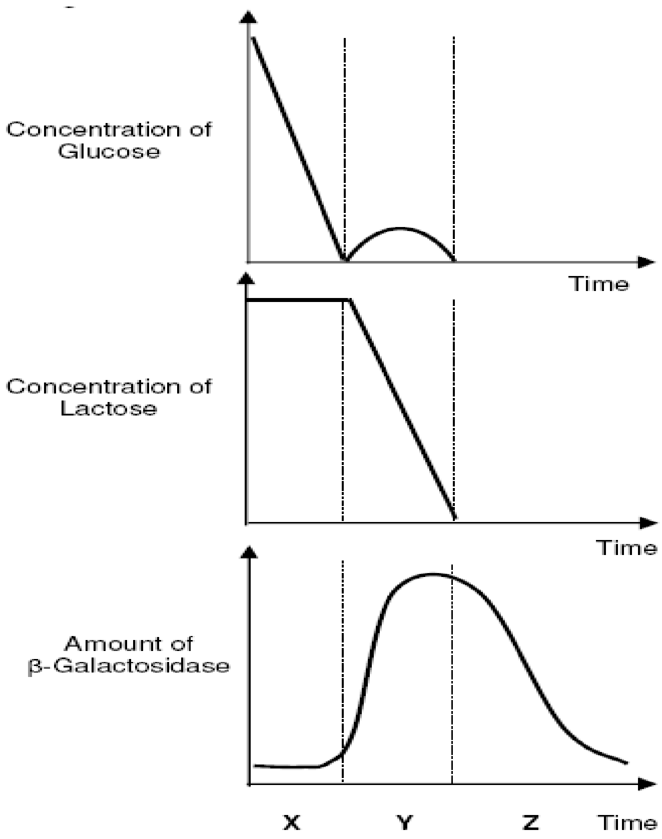
The above phage-infected bacteria are grown together with bacteria lacking the genes that code for enzymes required for synthesis of histidine (*His* $^-$), arginine (*Arg* $^-$), and methionine (*Met* $^-$).

After 24 hours, the progenies of bacteria that were *His* $^-$, *Arg* $^-$, *Met* $^-$ were observed to be able to grow on medium lacking histidine but not on medium lacking arginine and/or methionine.

Which option show the correct matching between the gene and its location on the phage-infected bacterium chromosome?

	Location 1	**Location 2**	**Location 3**
A	*His*	*Met*	*Arg*
B	*His*	*Arg*	*Met*
C	*Arg*	*His*	*Met*
D	*Met*	*Arg*	*His*

16. An experiment was conducted to examine the effects of glucose and lactose on the levels of β-galactosidase in *E.coli*. Lactose and glucose were added to a culture of bacteria at the start of the experiment and the levels of each were measured at specific time intervals. The results are shown in graphs below.



Which of the following statement is **wrong**?

A Amount of β-galactosidase only increases in period **Y** because glucose has been utilised.

B Allolactose binds to the lac repressor, allowing it to assume an active configuration such that it can bind to the operator in period **X**.

C mRNA of β-galactosidase has been degraded by nucleases in period **Z**.

D The slight increase in concentration of glucose is due to the hydrolysis of lactose by β-galactosidase in period **Y**.

17. The table shows the differences in the average length in kilobases (kb) of a gene and of messenger RNA (mRNA) in the bacterium, *Escherichia* coli, and a fruit fly, *Drosophila melanogaster.*

Organism	**Average length of gene (kb)**	**Average length of mRNA (kb)**
Escherichia coli	1.0	1.0
Drosophila melanogaster	7.2	3.0

Which statement **does not** help to account for the differences between the two organisms?

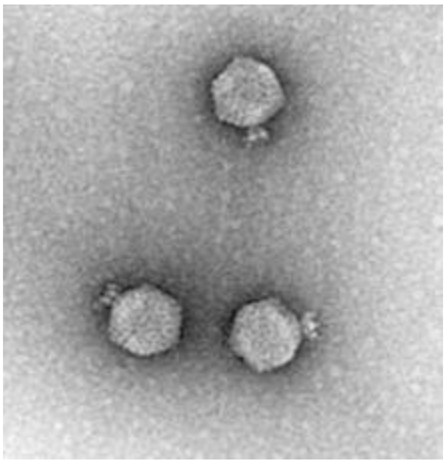
A The fruit fly genome is much larger than the bacterial genome.

B Fruit fly proteins are larger, on average, than the proteins of prokaryotes.

C Initial transcripts of pre-mRNA are shortened by splicing in fruit flies.

D Most fruit fly genes are interrupted by introns.

18. The following shows an electron micrograph of enterobacteria phage P22. It infects the bacterium *Salmonella typhimurium*. Its genetic material and replication cycle is similar to that of lambda phage.



Which of the following is most likely true about enterobacteria phage P22?

A It lacks tail fibres and enters *Salmonella typhimurium* by membrane fusion.

B It can be used to facilitate transfer of specific genes between *Salmonella typhimurium*.

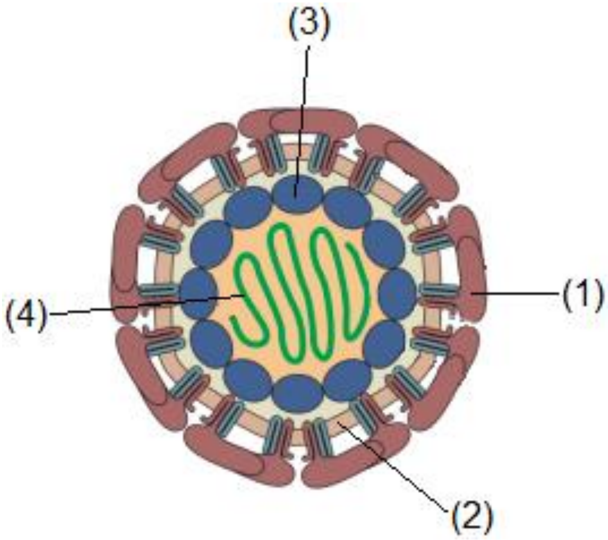
C It contains reverse transcriptase which will convert its RNA genome into double stranded DNA to be integrated into *Salmonella typhimurium* genome.

D Its linear genome circularises into a circular plasmid in the cytoplasm of *Salmonella typhimurium* and is passed down to daughter *Salmonella typhimurium* cells during binary fission.

19. Zika virus (ZIKV) is a member of the virus family Flaviviridae. It is spread by daytime-active Aedes mosquitoes, such as A. *aegypti* and A. *albopictus*.

ZIKV has recently been associated with microcephaly in foetuses, and with Guillian–Barré syndrome in adults. Recent research into its structures and replication cycle has aided medical scientists in designing potential vaccines against the virus.

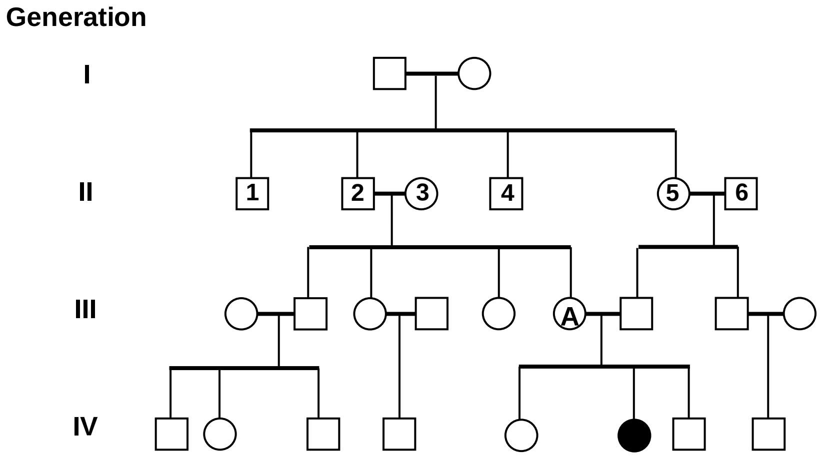
The diagram below shows the structure of ZIKV.



Which of the following correctly shows the components of each labelled structure in the diagram?

	1	2	3	4
A	Lipids	Phospholipids	Protein	DNA
B	Protein and carbohydrates	Phospholipids and cholesterol	Protein and carbohydrates	DNA
C	Protein and carbohydrates	Phospholipids and cholesterol	Protein	RNA
D	Carbohydrates	Phospholipids	Protein and carbohydrates	RNA

20. Pedigree analysis is useful in the analysis of genetic traits. The figure below shows the occurrence of a genetic disease in a family over 4 generations.



Given that individual A is pregnant again, what is the probability that the new born is affected or is a carrier?

	affected	carrier
A	¼	½
B	¼	¼
C	½	¼
D	¾	½

21. A gene is found to control a rare disease, Wiskott–Aldrich syndrome (WAS) which is characterised by eczema, low platelet count and immune deficiency.

Another gene that controls hair type has 2 alleles, 1 allele results in straight hair and another codes for curly hair. Presence of both results in wavy hair.

In couple 1, a normal female with straight hair married a wavy-haired male suffering from WAS. Predicted phenotypic ratio of their offspring is as follows:

	WAS	Hair
Female	All normal	1 wavy : 1 straight
Male	All normal	1 wavy : 1 straight

For couple 2, a wavy-haired female suffering from WAS married a wavy-haired normal man. Predicted phenotypic ratio of their offspring is as follows:

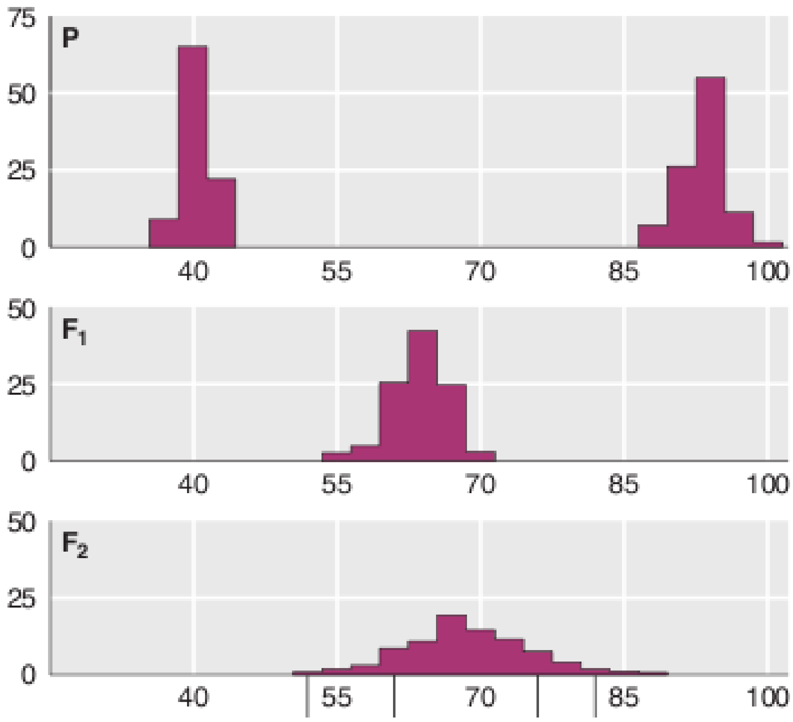
	WAS	Hair
Female	All normal	1 curly : 2 wavy : 1 straight
male	All affected	1 curly : 2 wavy : 1 straight

Which option is true regarding the inheritance of the two traits?

	WAS disease	**Hair type**
A	Autosomal	Codominance
B	Sex-linked	Incomplete dominance
C	Sex-linked	Codominance
D	Autosomal	Incomplete dominance

22. Two different pure-breeding varieties of cherry plants (P), which produce fruits of different sizes were crossed to produce the first generation (F1) offspring. Selfing of the F1 generation produced the second generation (F2) offspring.

The graphs below show the variation in fruit size in the parental (P) varieties, the first generation (F1) and the second generation (F2) of offspring.



Which of the following statement is true?

A The inheritance shows a continuous variation where the environment has small effect on the fruit size of the plants.

B The variation in the parental variety is due to the influence of one gene and environment.

C The difference in the variation between the F1 generation and F2 generations is a result of environmental influence only.

D When crossed further, the F3 generation will show greater variation due to presence of recombinant phenotypes.

23. The following statements describe the ATP synthase.

- ATP synthase is responsible for the synthesis of more than 90% of the ATP produced by mammalian cells under aerobic respiration.
- ATP synthase is a large enzyme with a water soluble complex (F_1) and a membrane complex (F_0).
- The F_1 complex contains three active sites.
- Rotation of the rotor in F_0 causes conformational change in the active sites in F_1, driving the phosphorylation of ADP to ATP.

Which of the following **cannot** be concluded from the above statements?

A Part of the ATP synthase is embedded in the mitochondrial membrane while part of the ATP synthase is in the matrix.
B ATP synthase is a complex molecule composed of multiple polypeptides.
C With three active sites, ATP synthase can produce 3 ATP when H^+ moves across the inner mitochondrial membrane.
D The F_0 complex facilitates the movement of H^+ from the intermembrane space to mitochondrial matrix.

24. The following are statements made regarding respiration.

- Oxygen is utilized in the cytosol during aerobic respiration.
- Carbon dioxide is released in the cytosol during aerobic respiration.
- Water is utilized in the mitochondria during aerobic respiration.
- Carbon dioxide is released during lactate fermentation.
- NAD+ is regenerated during fermentation so that ATP formation can still occur in anaerobic respiration.

How many statement(s) is/are correct?

A 1
B 2
C 3
D 4

25. The following are some statements regarding classification and phylogeny.

1 The binomial nomenclature is a naming system that helps to differentiate species.

2 Classifying organisms based on shared characteristics does not establish evolutionary relationship between organisms.

3 In phylogenetic tree, relationship between common ancestor and the different species is essential in showing evolutionary relationship.

4 Using molecular method to construct phylogenetic tree is subjective and depends on age of species.

5 A shared derived character is a unique character of a group of organisms not found in the common ancestor.

Which statements are true?

A 1, 2 and 3 only

B 2, 3 and 4 only

C 3, 4 and 5 only

D 1, 2, 3 and 5 only

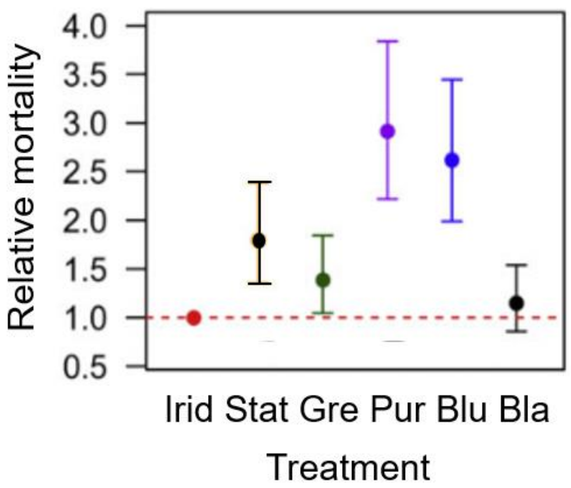
26. Some beetles display iridescence where its outer covering is shiny when viewed from different angles. In the Asian jewel beetles, *Sternocera aequisignatal,* their iridescent wing cases appear as a shiny mix of green, blue, purple and black depending on the viewing angles.

In a study, researchers took empty wing cases and coloured them. The empty wing cases are used to mimic the appearance of beetles. Some of the wing cases were iridescent (Irid), while other wing cases were coloured with nail polish that matched the colours on the iridescent wing cases. The wing cases were of the following treatments:

- iridescent (Irid)
- combination of green, purple, blue and black (Stat)
- green (Gre)
- purple (Pur)
- blue (Blu)
- black (Bla)

For the Stat treatment, the colours do not change with differing viewing angles unlike the iridescent wing cases.

Birds were then allowed to prey on the wing cases. The number of wing cases of the different treatments preyed by birds was compared to the iridescent wing cases. This relative mortality value may be used to indicate the relative mortality of the beetles. The results are shown in the following graph.



Which statement best explains the results of the investigation?

A Beetles with black cases help to camouflage and reduce probability of being seen by predators.

B Beetles with purple cases is likely to have highest reproduction rate.

C Iridescence helps the beetles increase the probability of finding their mates.

D The results for beetles with black cases is the least reliable.

27. The Homerus Swallowtail, a species of butterfly, is once abundant in the forests of Jamaica. They are now critically endangered due to significant pressures from habitat loss, predation and over-collecting.

Historical studies showed that there were three main populations of the Swallowtail in Jamaica, namely the Eastern, the Central and the Western populations. Each region had its own distinct elevation, humidity and host plants significantly different from the other regions.

Since 1925, there has been no Swallowtail sighted from the Central population. Logging and the expansion of agriculture resulted in extensive habitat destruction which might have resulted in the extinction of this population.

Scientists hypothesised that the mating of individuals between the Central population and peripheral populations might have served as a means for the sharing of genes.

The Eastern population has an estimated population size of less than 200 individuals and the Western population has an estimated population size of less than 50 individuals.

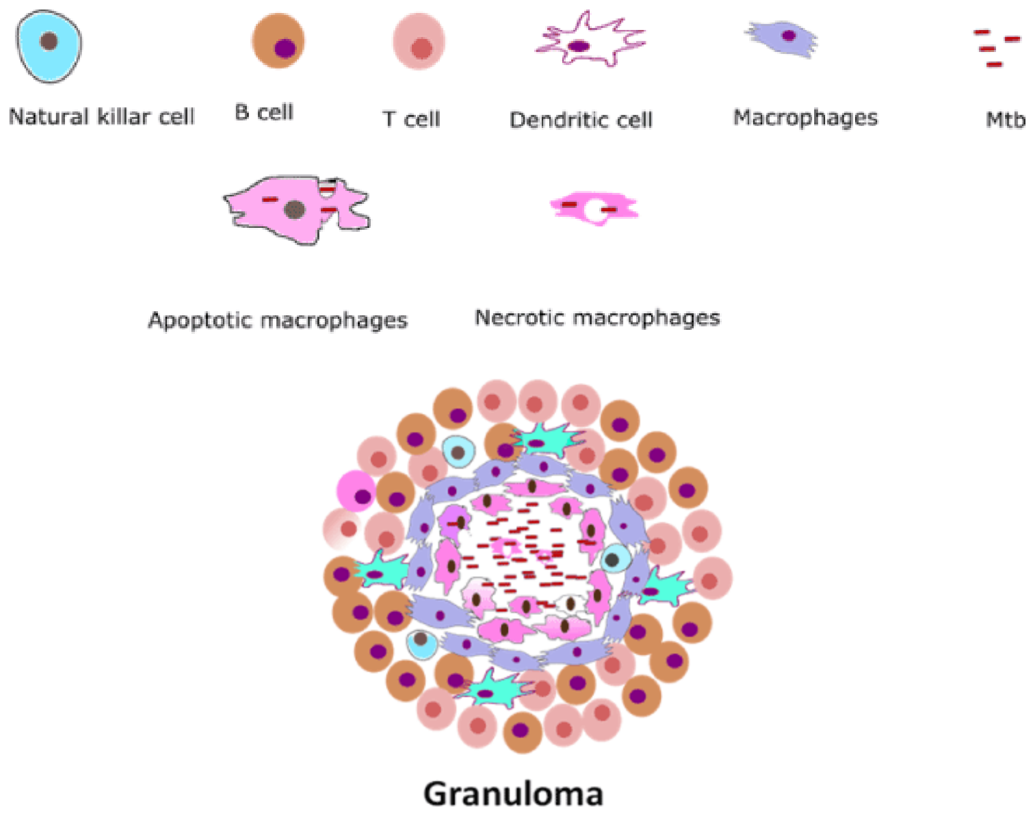
In the last twenty years, due to the effects of climate change, the Eastern population has been faced with significant decrease in rainfall while the Western population has experienced a marginal increase in rainfall.

Using modelling studies, scientists have predicted that with time, it is possible for Swallowtail on Jamaica to evolve into different species.

Which of the following best describes the evolution of the Homerus Swallowtail in Jamaica?

	genetic variability	type of speciation
A	low variability	allopatric
B	low variability	sympatric
C	high variability	allopatric
D	high variability	sympatric

28. The granuloma is an organised structured formed during infection of *M.tuberculosis* (Mtb). The structure composed of a few immune cells surrounding the *M.tuberculosis*.



Which are true about *M.tuberculosis* and formation of granuloma?

1 The structure comprises only of cells from the adaptive immune response.

2 Granuloma is a complex structured that acts to provide shelter for *M.tuberculosis.*

3 Cytokine can be secreted by the immune cells within the granuloma to stabilise the structure.

4 *M.tuberculosis* is a rod-shaped bacteria that causes air-borne disease.

5 Vaccine is readily available to tackle *M.tuberculosis* infection.

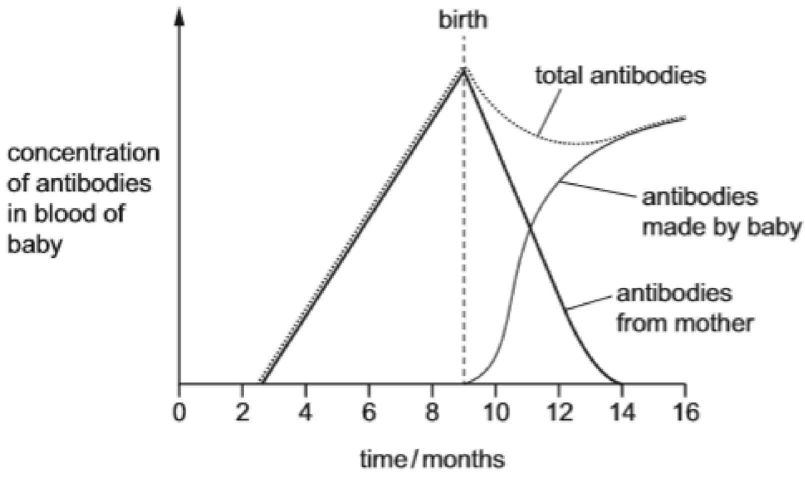
A 1, 2 and 3 only

B 1, 4 and 5 only

C 2, 3 and 4 only

D 3, 4, and 5 only

29. The graph shows the changes that occur in the concentration of antibodies in the blood of a baby before birth and during the first few months after birth.



Which description about the changes in immunity during the first few months after birth is correct?

A Active artificial immunity decreases, active natural immunity increases

B Active natural immunity decreases, active artificial immunity increases

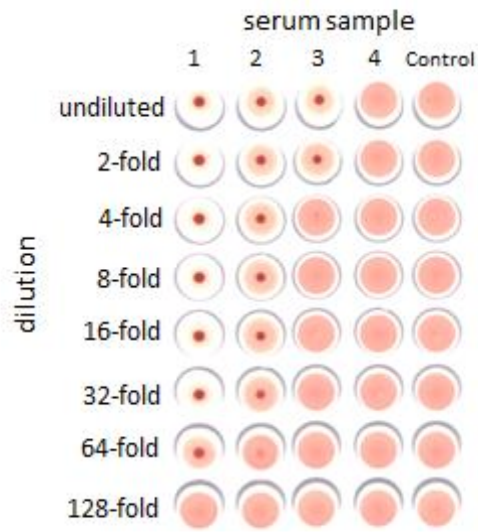
C Passive artificial immunity decreases, active natural immunity increases

D Passive natural immunity decreases, active natural immunity increases

30. The effectiveness of a vaccine is measured by the concentration of antibodies produced that can bind to specific antigens in the vaccine.

Blood from a man is taken at different times after vaccination and the blood serum containing antibodies is extracted. The serum is then subjected to a serial dilution where the concentration is halved with each step of the dilution. Each diluted blood serum is added to a well containing a fixed concentration of antigens. At a sufficiently high antibody concentration, a visible dark spot is formed due to a high level of antigen-antibody agglutination.

The diagram shows the results of an investigation on the effectiveness of the initial dose of the vaccine and its booster given after six months. Four serum samples were taken at different time points.



Which serum samples correspond to the different time points after the initial dose and the booster were given?

	2^{nd} day after initial dose is given	Before booster is given (6 months after initial dose)	2^{nd} day after booster is given	6 months after booster is given
A	1	2	4	3
B	4	2	3	1
C	4	3	1	2
D	1	4	2	3

End of Paper